Cuba Water Resources | Data Guide for supporting documents 1 thru 7

Objective

- Produce a small, defensible set of plots and maps addressing the project's research questions (bottom of page).

- The available materials are strongest for Q1 (trends at sampled wells), Q2 (well characteristics), Q4 (regional aquifer comparison), Q7 (local stream observations), Q8 (coverage/gaps), and Q9 (dashboard context). None independently establishes depletion of Cuba's municipal aquifer or identifies its current producing formation. Distinguish Cuba-local evidence from countywide and regional context in every caption.

What to expect from each support document

Support brief	Best use	Suggested output / caution
1 · USGS Sandoval groundwater statistics	9 wells / 9,462 aggregate records; Q1, Q8, Q9	Annual mean depth-to-water by well; trend slopes and monitoring coverage. No Cuba wells.
2 · NMBGMR Sandoval groundwater levels	87 wells / 456 records; Q2, Q8, Q1	Map unique wells; depth-to-water vs date for repeat-observed wells; well-depth comparison. Closest reported well ~24 mi from Cuba.
3 · Aquifer sensitivity polygons	613 attribute rows; Q9 only if validated	Do not map from CSV alone. Obtain GIS geometry, CRS and definition of "sensitivity"; verify Cuba overlap.
4 · USGS Rio Puerco at Cuba	10 discrete flow visits; Q7, Q8, Q9	Scatter of flow (ft^3/s) vs visit date; mark 6 zero-flow observations. Local stream, not municipal groundwater.
5 · USGS Ojo Alamo Sandstone	Regional geological map plate; Q4, Q2	Extract/attribute aquifer depth, thickness and well-location maps. Not a numeric CSV.
6 · San Jose Formation near Lindrith	Regional study and maps; Q4, Q2	Use hydrostratigraphic section, Cuba Mesa thickness and head maps; compare cautiously with doc 5.
7 · OSE San Juan well logs	1,379 construction records; Q2, Q8	Depth histograms, first-water vs borehole-depth scatter, well map. Only 6 coded Sandoval; Cuba link unverified.

Research questions

1. How do groundwater levels vary across Sandoval County, and which areas show sustained decline, stability, or recovery?
2. What do well records show about well depth, depth to water, and the aquifer units that communities rely on?
3. Do the available data support the concern that Cuba's main groundwater source is approaching severe depletion?
4. How do the Ojo Alamo Sandstone and San Jose Formation compare near Cuba in extent, thickness, hydraulic properties, water levels, and water quality?
5. How does groundwater quality, including salinity, vary across the county and near Cuba?
6. Where are acequias located, what surface water do they depend on, and which communities do they serve?
7. Do changes in groundwater levels line up with nearby surface-water or acequia conditions, or are the records too limited to tell?
8. Where is monitoring coverage strong, and where are the most important data gaps?
9. Which indicators can be combined into a dashboard that shows water-resource stress without overstating how certain the data are?